



Seed Amplification Assay of Formalin-Fixed Skin Biopsies

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Summary

The seed amplification assay (SAA) has emerged as a highly sensitive and specific assay for detecting pathological alpha-synuclein (α Syn) in biomatrices such as cerebral spinal fluid (CSF), skin, and the submandibular gland (SMG) to diagnose Parkinson's disease (PD) and related disorders. To date, most studies have used CSF to seed the assay, however, the spinal tap is an invasive procedure and peripheral skin tissue may have equal diagnostic power while remaining noninvasive. Therefore, our study sought to establish the clinical utility of the SAA using cryosectioned and formalin-fixed skin biopsies from PD and healthy control (HC) patients in the PPMI cohort. We received blinded cryosectioned and formalin-fixed samples from 31 patients and unblinded samples from 3 PD and 3 control patients. Based on our optimization studies, we found cryosectioned samples are technically challenging to use in the SAA because of OCT contamination and the small amount of tissue per sample provided. Through the development and optimization of our formalin-fixed skin processing, we recovered sufficient seeding activity. However, the low sensitivity (~60%) and specificity (~60%) of these samples in our study may indicate the need for either further assay optimization or additional samples from each patient.

Method

Protein Purification

Our protein purification protocol was adopted from Groveman et al. [1] with minor modifications. Briefly, WT α Syn was expressed in BL21 *E. coli* transfected with a pT7-7 expression plasmid. The bacterial mini culture was grown at 37°C with 230 rpm shaking for 8 h. The mini culture was transferred to pre-warmed 1 L LB broth with antibiotics and Overnight Autoinduction reagents. Once the OD₆₀₀ reached 1.2 – 1.3, the 1-L culture was divided equally into 4 X 250-mL bottles. The bacterial cells were pelleted by centrifugation for 20 min at 3,750 rpm at 4°C. The supernatant was discarded. Each pellet was resuspended in 25 mL of osmotic shock buffer and rocked gently at room temperature for 10 min. Tubes were centrifuged at 9,000 x g at 20°C for 20 min. The supernatant was discarded, and pellets were placed on ice. Each pellet was gently resuspended with a serological pipet in 25 mL of approximately 2 mM MgCl₂ and incubated for 5 min at 4°C with gentle rocking. The tubes were centrifuged at 9,000 x g at 4°C for 30 min. The supernatant was collected in a 150-mL sterile glass beaker with a magnetic stir bar and kept immersed in ice. The beaker was placed on a stirrer and gently stirred as the pH was reduced to 3.5 using 1 M HCl





(a white precipitate appears, turning the solution white and cloudy) and continued stirring for 5 min. The solution contents were transferred into four 50-mL tubes and centrifuged at 9000 x g at 4°C for 30 min and the supernatant was collected in a 150-mL sterile glass beaker. The beaker was placed on a stirrer and the solution was gently stirred as the pH was increased to 7.5 using 1 M NaOH. Streptomycin sulfate was added to make a final concentration of 2.5 mg/mL and gently placed on a flat container set on a rocker allowing it to roll back and forth for 10 min at 4°C. The tubes were centrifuged for 20 min at 24,000 x g at 4°C and the supernatant was collected in two 50-mL tubes kept on ice. Before application to the FPLC column, the sample was concentrated from 100 mL to 15-20 mL using 3-kDa centrifugal filters at 4,000 x g. The precipitate at the bottom of the filter was resuspended and collected. Finally, the sample was filtered using a 0.22- μ m syringe filter.

Size-Exclusion Chromatography (SEC)

Next, 10 mL of the prepared sample was manually loaded into the 10-mL sample loop of the Bio-Rad NGC chromatography system and systemically loaded into the column using 15 mL of the SEC buffer. At a flow rate of 1 mL/min, 2 column volumes (CV) of fresh SEC buffer was used to elute the target protein in 3-mL fractions. The fractions with a high absorbance peak were visualized using SDS page and Coomassie staining. Fractions with strong putative α Syn expression (bands around 14 kDa) were pooled and carried into anion-exchange chromatography.

Anion-Exchange Chromatography

Following SEC, 10 mL of the pooled fractions from size-exclusion chromatography was manually loaded into the NGC chromatography system and subsequently loaded into the anion-exchange column (HiPrep Q FF 16/10) using 15 mL of low-salt buffer. At a flow rate of 1 mL/min (the isocratic segment), 2 CV of 5% high-salt buffer was pushed through the column without fraction collection. This was followed by a fraction collection elution stage using a flow rate of 1 mL/min with a 6-CV gradient from 20% to 55% of high salt buffer with the collection of 2-mL fractions. Fractions under the absorbance peak were collected for SDS-PAGE analysis. The Coomassie-stained gel confirmed the expression of α Syn. Fractions with strong α Syn expression were pooled and carried into dialysis. Aliquots were made in 1.5-mL low protein-binding tubes and stored at -80°C. Schematic 1 shows a detailed graphic of the full in-house α Syn purification method.

Formalin-Fixed Skin Biopsy Homogenate Preparation & Optimization

The formalin-fixed skin processing was adopted from our previously published protocol [2] with changes made to accommodate the difference between FFPE sections and formalin-fixed skin





biopsies. In preliminary experiments, we used our published protocol, but identified insufficient amplification of the PD samples. Briefly, 950 μ L of 500 mM tris + 0.02% SDS buffer (TSB, pH 8.0) was added and mixed for 1 min. The samples went through 3 freeze-thaw cycles. The samples were again mixed for 10 min followed by centrifugation at 25°C for 5 min (13,000 x g) and the supernatant was discarded. Fresh 1 x TSB was then added and mixed for 10 min using Intelli-Mixer. Samples were centrifuged at 25°C for 5 min again (13,000 x g) and supernatant was removed while retaining about 90 μ L of buffer. The samples were placed in the ThermoMixer at 1,500 rpm and 90°C for 20 min. The temperature was reduced to 60°C for an additional 90 min and continued 1,500-rpm shaking. Finally, samples were sonicated for 8 min at 90% with cycles of 30-s intervals of sonication and 30 s of rest for a total sonication time of 4 min.

Following a series of optimization experiments using formalin-fixed scalp autopsy samples, we identified the following tissue processing protocol. The skin biopsy was weighed, minced, and suspended in 10% w/v of fresh TSB. The skin was incubated for 4 h at 90°C with 500-rpm shaking. Following incubation, the buffer was removed and washed three times with 1 x TBS. Collagenase A (0.25%) and CaCl₂ (2 mM) were added to make a 10% w/v and zirconium oxide beads were added. Samples were incubated for 4 h with 500-rpm shaking. Samples were homogenized for 5 min using a bead beater with a 1-min rest on ice at the halfway point (2:30). Samples were sonicated for 4 min at 75% with cycles of 30-s intervals of sonication and 30 s of rest for a total sonication time of 2 min. The homogenate was centrifuged at 5,000 rpm for 5 min, and the supernatant was collected and stored at -80°C.

Cryosectioned Skin Biopsy Homogenate Preparation

Similar to formalin-fixed skin, we found the OCT in cryosections impaired the SAA aggregation reaction and thus required optimization. OCT-embedded skin sections stored in cryovials were suspended in 500 μ L 1x TBS and transferred to a new tube. The samples were incubated with shaking on the Intellimixer for 5 min and then centrifuged for 5 min at 5,000 x g. The supernatant was removed, and this wash step was repeated 4 times. The pellet was resuspended in 100 μ L of 1x TBS, 2 mM CaCl₂, 0.25% Collagenase A and incubated for 4 h at 37°C and 450 rpm. Zirconium oxide beads were added, and a bead beater was used for 1 min. The samples were stored at -80°C.

Seed Amplification (SAA)

The SAA method has been described in our prior publications [2–4]. A 96-well clear-bottom plate (Thermo Fisher Cat# 1256672) was loaded with six 800-um silica beads (OPS Diagnostics Cat# 80020002) in each well. Purified α Syn monomer was thawed from -80°C and centrifuge-filtered (100 kDa) to remove aggregates that may have formed in storage. The reaction mixture contained 0.1 mg/mL α Syn monomer (as measured by the NanoDrop), 10 μ M ThT, 40 mM dibasic sodium phosphate buffer (pH 8.0), 0.00125% SDS, and varying concentrations of NaCl depending on the





experimentally determined optimized concentration. Next, 98 μ L of the reaction mixture was loaded into each well of the plate followed by 2 μ L of biomatrix from the specified dilution. The plates were sealed with a plate sealer (Thermo Fisher Cat# 235307) and incubated in the CLARIOstar microplate reader (BMG Labtech) at 42°C. The reader settings were as follows: double orbital shaking at 400 rpm for 1 min followed by 1-min resting incubation between shaking cycles; the gain setting was between 1,600 and 1,800; ThT fluorescence was measured every 30 min for a minimum of 42 h using 445 nm excitation and 480 nm emission wavelengths. Samples were run as quadruplicates unless otherwise specified. The kinetic parameters evaluated were protein aggregation rate (PAR), fluorescent maximum (Fmax), and hit rate. The threshold line used to calculate PAR and hit rate is defined as the average of the first ten readings of all samples plus ten standard deviations (SD).

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